

Reference point for the slow landslide time series

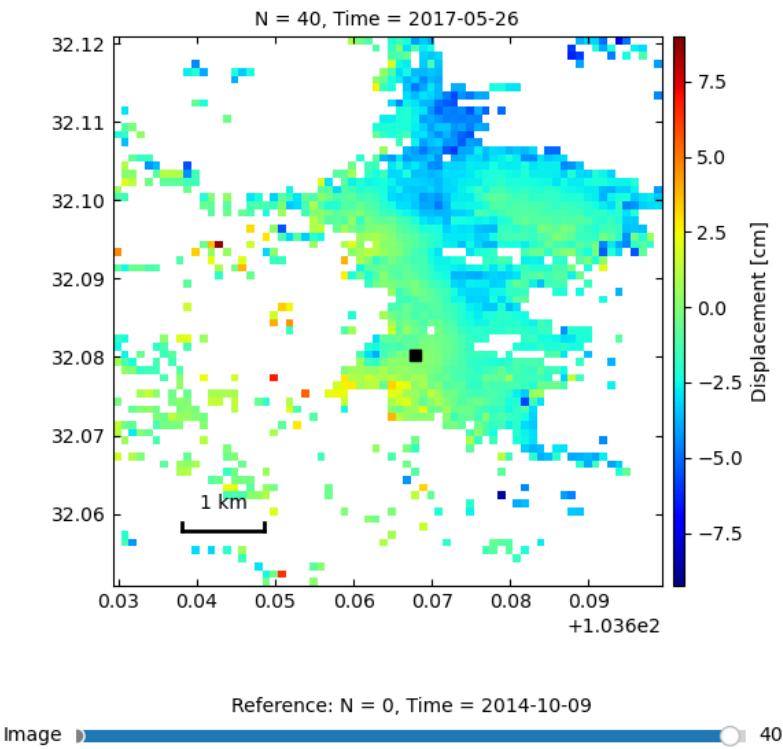


Fig. 1 Displacement Map

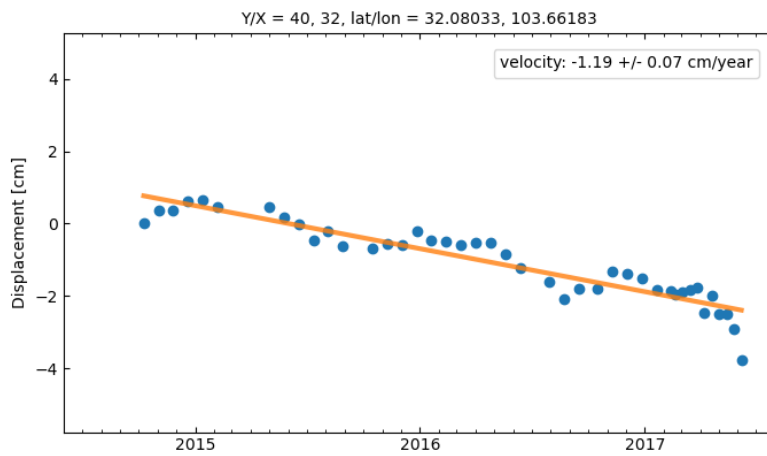


Fig. 2 Point Displacement Timeseries

Phase coherence is an effective tool to map the landslide because the sudden surface disturbance completely destroys the phase correlation. However, it can suffer from false positives due to natural low coherence in heavily forested surroundings.

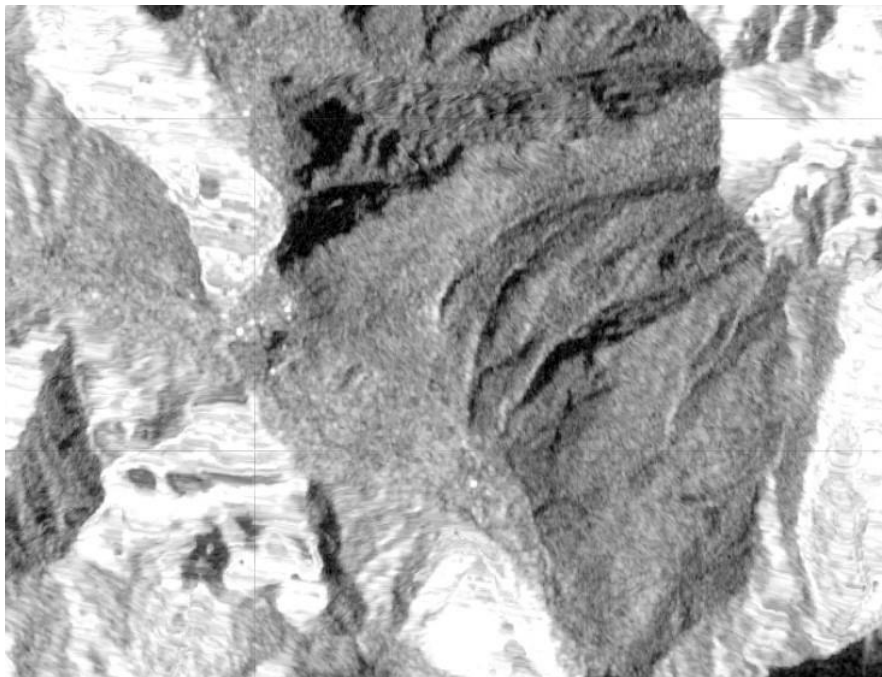


Fig. 3 Sentinel 1 - before landslide

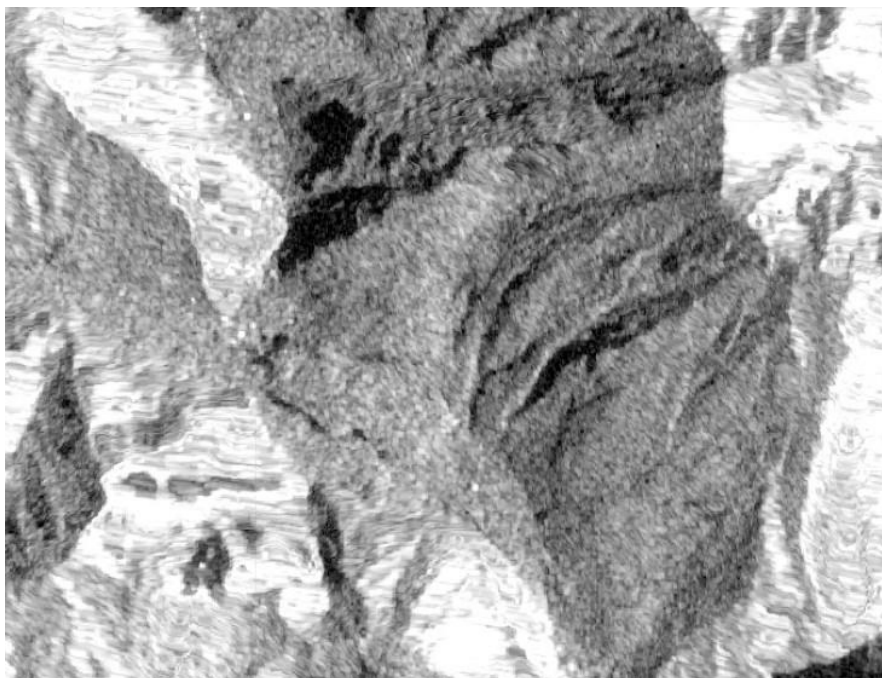


Fig 4. Sentinel 1 - after landslide

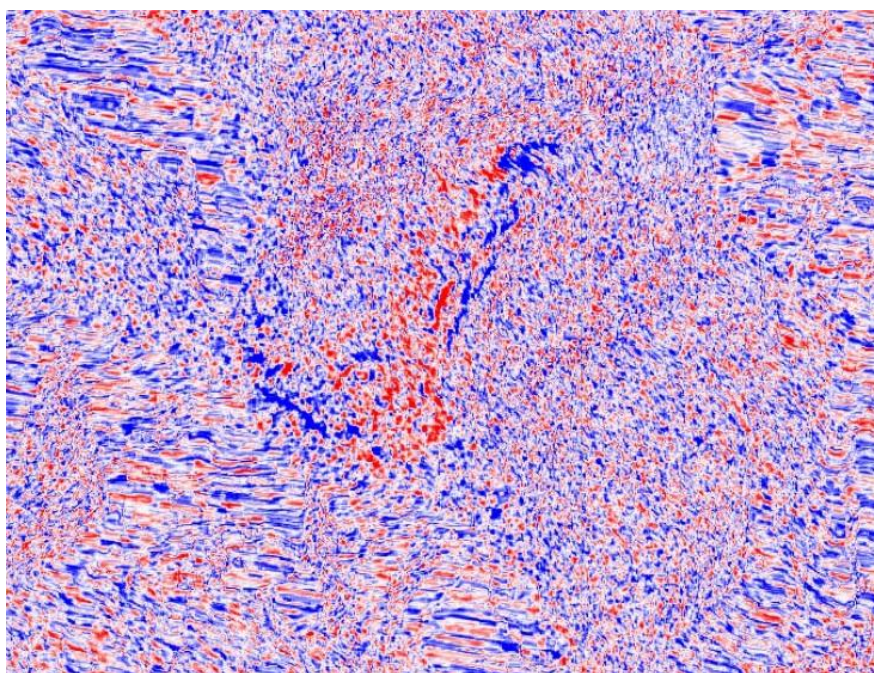


Fig. 5 Sentinel 1 – difference

Comparing the change in Sentinel-1 VV backscatter before and after the event acts as a powerful complement, as the distinct backscatter intensity shift provides a boundary of the landslide scar against the vegetation.



Fig. 6 Sentinel 2 - before landslide

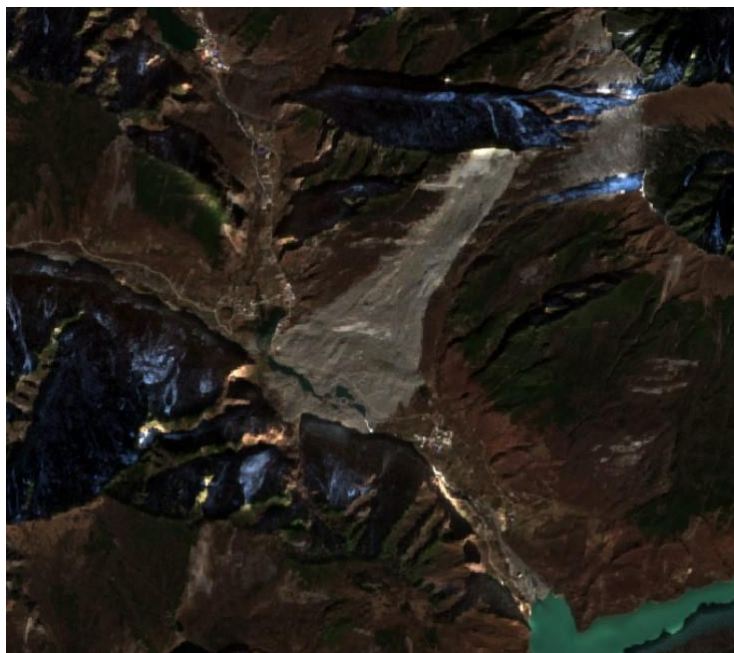


Fig. 7 Sentinel 2 - after landslide

Pre- and post-event Sentinel-2 imagery isolates the geographical extent of the landslide. The post-event image clearly reveals the massive grey path of exposed bare rock and debris cutting through the landscape.



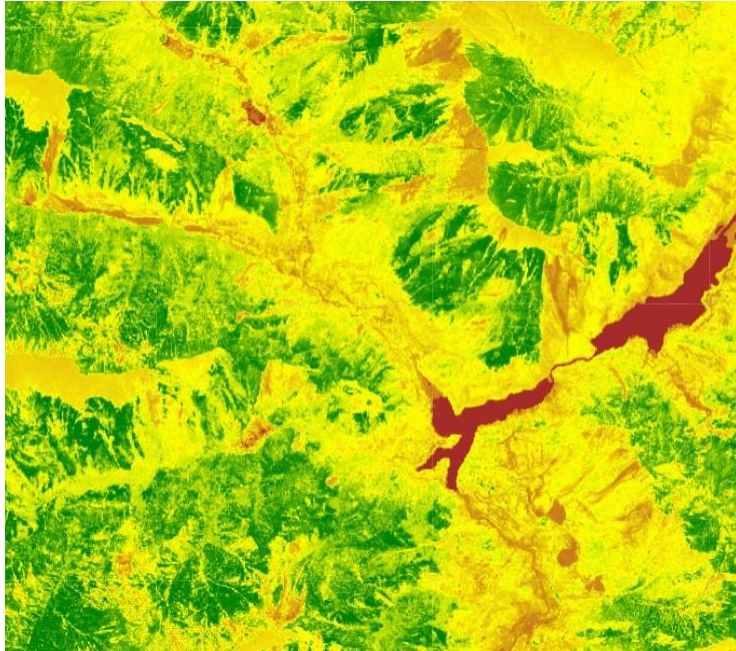


Fig. 8 Sentinel 2 NDVI - before landslide

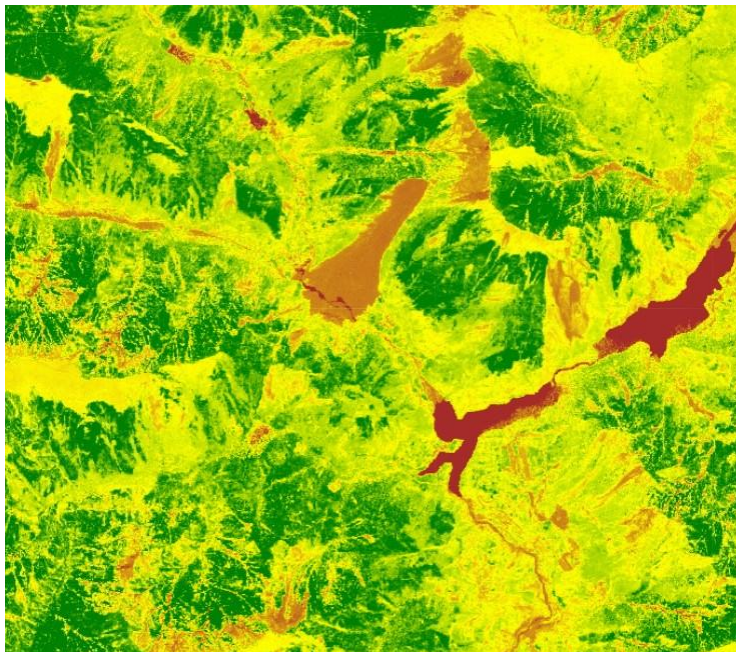


Fig. 9 Sentinel 2 NDVI - after landslide

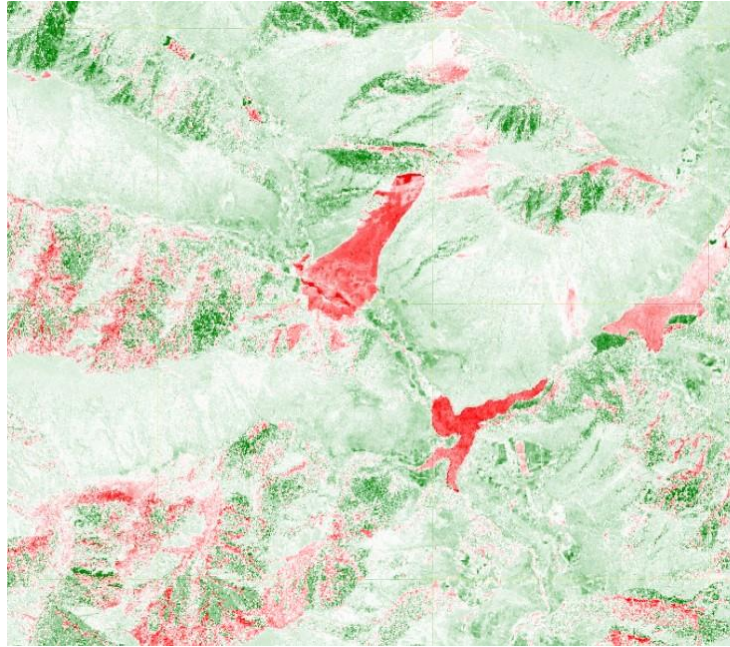


Fig. 10 Sentinel 2 NDVI - difference

The Normalized Difference Vegetation Index (NDVI) was computed to evaluate environmental impact. The NDVI difference map highlights a dramatic drop in values directly over the landslide track, successfully quantifying a catastrophic localized loss of vegetation and transition to barren land.